1. Login and add Students

```
cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade % /usr/local/bin/python3 /Users/cliffgonsalves/Docum
ents/Programming-For-DA/CheckMyGrade/main.py

Login Menu
1 Login as Student
2 Login as Professor
3 Register
4 Admin Login
0 Exit
Choice: 4
User ID: admin
Password: admin
Logged in as admin

CheckMyGrade System
1 Add Student
2 Display Students
3 Save Students
4 Modify Student
5 Manage Courses
6 Manage Professors
7 Import/Export All CSVs
8 Delete Student
0 Exit
Choice: 1
Email: ron@yahoo.com
First name: ron
Last name: swanson
Course ID: cs-113
Grade: C
Marks: 70
```

2. Modify, Delete, and demonstrated deleted student is not found

```
cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade % /usr/local/bin/python3 /Users/cliffgonsalves/Docum
ents/Programming-For-DA/CheckMyGrade/main.py

CheckMyGrade System
1 Add Student
2 Display Students
3 Save Students
4 Modify Student
5 Manage Courses
6 Manage Professors
7 Import/Export All CSVs
8 Delete Student
0 Exit
Choice: 4
Email of student to modify: ron@yahoo.com
Press Enter to keep the current value.
First name (ron):
Last name (swanson):
Course ID (cs-113): cs-116
Grade (C): B
Marks (70): 80
Student updated

CheckMyGrade System
1 Add Student
2 Display Students
3 Save Students
4 Modify Student
5 Manage Courses
6 Manage Professors
7 Import/Export All CSVs
8 Delete Student
0 Exit
Choice: 8
Email: ron@yahoo.com
Student deleted

CheckMyGrade System
1 Add Student
2 Display Students
3 Save Students
4 Modify Student
5 Manage Courses
6 Manage Professors
7 Import/Export All CSVs
8 Delete Student
0 Exit
Choice: 8
Email: ron@yahoo.com
Student not found
```

3.display sorted data based on student grades and searching through student records to find a student, and displaying the total time taken to find a student record

```
Login Menu
1 Login as Student
2 Login as Professor
3 Register
4 Admin Login
0 Exit
Choice: 2
User ID: charles@gmail.com
Password: charles123
Logged in as professor

Professor Menu
1 View Sorted Results by Grade (Course)
2 Search Student Records
3 Course Statistics (Average/Median)
4 Generate Grade Reports
5 Import/Export All CSVs
0 Exit
Choice: 1
Course ID: cs-110
creedb@aol.net, Creed, Bratton, cs-110, A, 98
cliff@gmail.com, Cliff, Gonsalves, cs-110, A, 90
mike@aol.net, mike, Scott, cs-110, C, 60

Professor Menu
1 View Sorted Results by Grade (Course)
2 Search Student Records
3 Course Statistics (Average/Median)
4 Generate Grade Reports
5 Import/Export All CSVs
0 Exit
Choice: 2
Student email to search: creedb@aol.net
creedb@aol.net, Creed, Bratton, cs-110, A, 98
Search time: 0.039 ms
```

4. Demonstrating Course Statistics (Average/Median)

```
Professor Menu
1 View Sorted Results by Grade (Course)
2 Search Student Records
3 Course Statistics (Average/Median)
4 Generate Grade Reports
5 Import/Export All CSVs
0 Exit
Choice: 3
Course ID: cs-110
Average score: 82.67
Median score: 90.00
```

5. Report by course

lent.py | charlesreport | csreport | course.py

csreport

```
Course Report: cs-110
creedb@aol.net, Creed, Bratton, cs-110, A, 98
cliff@gmail.com, Cliff, Gonsalves, cs-110, A, 90
mike@aol.net, mike, Scott, cs-110, C, 60
```

Debug Console (⇧⌘Y)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade % /usr/local/bin/py
ents/Programming-For-DA/CheckMyGrade/main.py
Professor Menu
1 View Sorted Results by Grade (Course)
2 Search Student Records
3 Course Statistics (Average/Median)
4 Generate Grade Reports
5 Import/Export All CSVs
0 Exit
Choice: 4

Report Menu
1 Report by Course
2 Report by Professor
3 Report by Student
0 Back
Choice: 1
Course ID: cs-110

Course Report: cs-110
creedb@aol.net, Creed, Bratton, cs-110, A, 98
cliff@gmail.com, Cliff, Gonsalves, cs-110, A, 90
mike@aol.net, mike, Scott, cs-110, C, 60
Save report to file? (y/n): y
Filename to save course report: csreport
Report saved to csreport
```

6. Report by professor

charlesreport

```
Professor Report: charles@gmail.com Charles
Course: cs-110 Rank: Senior Professor
creedb@aol.net, Creed, Bratton, cs-110, A, 98
cliff@gmail.com, Cliff, Gonsalves, cs-110, A, 90
mike@aol.net, mike, Scott, cs-110, C, 60
```

Debug Console (⇧⌘Y)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade % /usr/local/bi
ents/Programming-For-DA/CheckMyGrade/main.py
Report saved to csreport

Report Menu
```

7. Report by student

```
cliffreport
1   Student Report: Cliff Gonsalves
2   Course: cs-110 Grade: A Marks: 90
3
```

Debug Console (⇧⌘Y)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade % /usr/local/bin/py
ents/Programming-For-DA/CheckMyGrade/main.py
Report saved to csreport

Report Menu
1 Report by Course
2 Report by Professor
3 Report by Student
0 Back
Choice: 2
Professor ID: charles@gmail.com

Professor Report: charles@gmail.com Charles
Course: cs-110 Rank: Senior Professor
creedb@aol.net, Creed, Bratton, cs-110, A, 98
cliff@gmail.com, Cliff, Gonsalves, cs-110, A, 90
mike@aol.net, mike, Scott, cs-110, C, 60
Save report to file? (y/n): y
Filename to save professor report: charlesreport
Report saved to charlesreport

Report Menu
1 Report by Course
2 Report by Professor
3 Report by Student
0 Back
Choice: 3
Student email: cliff@gmail.com

Student Report: Cliff Gonsalves
Course: cs-110 Grade: A Marks: 90
Save report to file? (y/n): y
Filename to save student report: cliffreport
Report saved to cliffreport
```

8. Code manages four csv files

login_user.py | queue_ds.py | main.py | students.csv

data > students.csv

```
email,first_name,last_name,course_id,grade,marks
cliff@gmail.com,Cliff,Gonsalves,cs-110,A,90
c@gmail,Charles,Gonsal,cs-112,A,90
joe@yahoo.com,Joe,Doe,eg-127,B,78
mike@aol.net,mike,Scott,cs-110,C,60
creedb@aol.net,Creed,Bratton,cs-110,A,98
```

login_user.py | queue_ds.py | main.py | Professor.csv

data > Professor.csv

```
Professor_id,Professor Name,Rank,Course.id
charles@gmail.com,Charles,Senior Professor,cs-110
```

login_user.py | queue_ds.py | main.py | Login.csv | linked_list.py

data > Login.csv

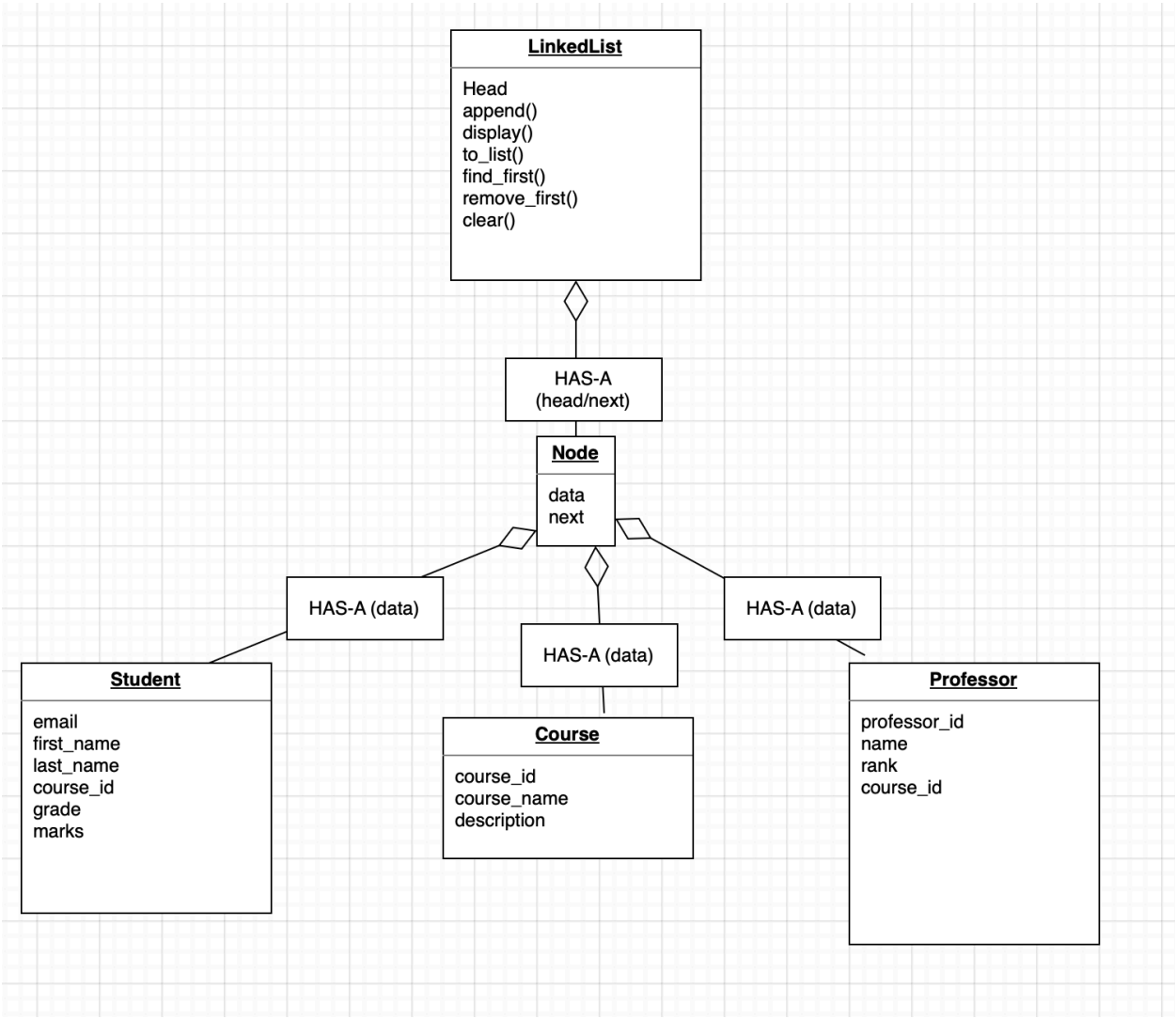
```
User_id,Password,role
cliff@gmail.com,6b9973a1f0ea8a3cd2bf7f5e2261da72e05a5bbc384bf51733a2c0c02d848b
charles@gmail.com,c65b65859229387d9e067f20cec12de6b9dfe1ea56fe9e68cc9bda902ee2
admin,8da193366e1554c08b2870c50f737b9587c3372b656151c4a96028af26f51334,admin
```

login_user.py | queue_ds.py | main.py | Course.csv | linked_list.py

data > Course.csv

```
Course_id,Course_name,Description
cs-110,Algorithms and data struct.,Intro to algos and Ds.
cs-112,distibuted systems,Intro to ds
eg-127,Engineering Management,Advanced management for engineering
```

9. IS-A/HAS-A relationship.



10. Checking for uniqueness and not null

```
● cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade % /usr/local/bin/python3 /Users/cliffgonsalves/Docum
  ents/Programming-For-DA/CheckMyGrade/main.py

  Login Menu
  1 Login as Student
  2 Login as Professor
  3 Register
  4 Admin Login
  0 Exit
  Choice: 4
  User ID: admin
  Password: admin
  Logged in as admin

  CheckMyGrade System
  1 Add Student
  2 Display Students
  3 Save Students
  4 Modify Student
  5 Manage Courses
  6 Manage Professors
  7 Import/Export All CSVs
  8 Delete Student
  0 Exit
  Choice: 1
  Email:
  Email is required, and cannot be empty.
  Email: cliff@gmail.com
  First name: cl
  Last name: g
  Course ID: abc-123
  Grade: A
  Marks: 99

  CheckMyGrade System
  1 Add Student
  2 Display Students
  3 Save Students
  4 Modify Student
  5 Manage Courses
  6 Manage Professors
  7 Import/Export All CSVs
  8 Delete Student
  0 Exit
  Choice: 3
  Error: duplicate email found (cliff@gmail.com). Not saving.
```

11. Passwords are encrypted and not saved as is

data > Login.csv

```
User_id,Password,role
cliff@gmail.com,6b9973a1f0ea8a3cd2bf7f5e2261da72e05a5bbc384bf51733a2c0c02d848b
charles@gmail.com,c65b65859229387d9e067f20cec12de6b9dfe1ea56fe9e68cc9bda902ee2
admin,8da193366e1554c08b2870c50f737b9587c3372b656151c4a96028af26f51334,admin
```

5. Unit Tests on 1000+ records, Load the data saved in the csv files/ Load and search/ printing the total time taken in search cases, testing sorting based on marks, and Unit tests to add/ delete/modify the course, professor and students.

```
cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade % python3 -m unittest discover -s tests
Running test_add_delete_modify_course
.Running test_add_delete_modify_professor
.Running test_add_delete_modify_students_with_1000_records
.Running test_load_previous_csv_and_search_timing
Search time (CSV load): 0.001 ms
.Running test_save_and_load_1000_records
.Running test_sort_students_by_marks_and_email_with_timing
Sort by marks ascending: 0.062 ms
Sort by marks descending: 0.060 ms
Sort by email ascending: 0.035 ms
Sort by email descending: 0.029 ms
.
----------------------------------------------------------------------
Ran 6 tests in 0.013s

OK
cliffgonsalves@Cliffs-MacBook-Air CheckMyGrade %
```